

Employee Attrition Analysis

Statistical Theory Course

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Abstract

Employee attrition poses significant operational and financial challenges for organizations. This study investigates employee characteristics associated with attrition using the IBM HR Analytics dataset ($n = 1,470$). We combined False Discovery Rate (FDR)-corrected univariate tests with multivariable logistic regression. Univariate analysis revealed significant differences across compensation, career, and workplace factors. In the multivariable model, `OverTime` and frequent `BusinessTravel` were strongly associated with higher attrition odds, whereas longer tenure in the current role was protective. Notably, `MonthlyIncome` lost statistical significance after adjusting for confounding variables. Two hypothesized interactions (`OverTime` $\times$ `MonthlyIncome` and `DistanceFromHome` $\times$ `MaritalStatus`) did not improve model fit and were excluded. Evaluated on an independent test set, the final model achieved a ROC-AUC of 0.797. The findings underscore the necessity of multivariable approaches to isolate independent attrition drivers.

I. Introduction

Employee attrition, defined as the departure of employees from an organization, is an important topic in human-resource and organizational research. Understanding attrition is valuable because employee turnover can disrupt organizational continuity and requires organizations to replace lost knowledge and experience. More importantly from an analytical perspective, employee departure may be associated with measurable characteristics of the employee and the workplace. These characteristics include compensation, career experience, job responsibilities, working conditions, and demographic factors. Identifying such relationships can therefore contribute to a better understanding of which characteristics distinguish employees who leave from those who remain.

A substantial body of research has examined the mechanisms underlying employee turnover. Tett and Meyer [1] conducted a meta-analysis based on 178 independent samples from 155 studies to investigate the relationships among job satisfaction, organizational commitment, turnover intentions or withdrawal cognitions, and actual turnover. Their findings showed that job satisfaction and organizational commitment each contributed independently to turnover intentions, with job satisfaction

showing the stronger relationship. They further found that turnover intentions and withdrawal cognitions mediated much of the relationship between work attitudes and actual turnover. These findings established an important connection between employees' attitudes toward their work and their subsequent decisions to leave an organization.

As employee-level data have become increasingly available, attrition has also been approached as a statistical classification and prediction problem. Yedida et al. [2] investigated whether employee attrition could be predicted from measurable workplace characteristics using a k-nearest neighbors classifier. Their analysis incorporated variables including employee performance evaluation, average monthly working hours, and years spent at the company. Their results demonstrated that employee characteristics can be combined within a quantitative framework to distinguish between employees who remain and those who leave, illustrating the potential of data-driven methods for employee attrition analysis.

Particularly relevant to the present study, Yang and Islam [3] analyzed the IBM Employee Attrition dataset to identify characteristics associated with employee resignation. Their analysis combined correlation analysis, Random Forest feature selection, K-means clustering, and binary logistic regression. They identified variables such as monthly income, age, and the number of companies previously worked for as important characteristics in their analysis. Their logistic regression results further indicated substantially higher attrition among employees who traveled frequently compared with those who traveled rarely. This study demonstrates that the IBM dataset contains relationships involving compensation, career history, and workplace conditions that can be investigated using both exploratory and multivariable statistical methods.

The present study builds on this literature by examining employee attrition from complementary univariate and multivariable perspectives. We use the IBM HR Analytics Employee Attrition dataset, which contains 1,470 employee observations and includes demographic, compensation-related, career, and workplace characteristics. Rather than relying on a single statistical comparison, we first examine individual associations between employee characteristics and attrition and then use multivariable logistic regression to determine which relationships re-

main after accounting for other explanatory variables. We additionally examine whether selected relationships are better represented through interaction effects, allowing us to investigate whether the association of one characteristic with attrition depends on another characteristic.

Accordingly, the study addresses three main hypotheses. First, we examine whether the distributions of selected numerical employee characteristics differ between employees who leave and those who remain. Second, we examine whether categorical workplace and employee characteristics are associated with attrition status. Third, we examine whether selected employee characteristics retain significant associations with attrition when considered simultaneously in a multivariable logistic regression model. Together, these analyses aim to distinguish relationships that are apparent when characteristics are examined individually from those that remain independently associated with attrition after adjustment for other factors.

II. Results

The analysis addressed three main hypotheses concerning employee attrition. First, we examined whether selected numerical and ordinal employee characteristics differed between employees who left the organization and those who remained. Second, we examined whether categorical employee and workplace characteristics were associated with attrition. Third, we evaluated which characteristics remained independently associated with attrition when considered simultaneously in a multivariable logistic regression model. All univariate hypothesis tests were performed on the training set, and multiple comparisons were controlled using the Benjamini–Hochberg false discovery rate (FDR) procedure separately within the numerical and categorical test families.

A. Attrition Distribution

The complete dataset contained 1,470 employees, of whom 237 (16.1%) had left the organization and 1,233 (83.9%) remained. Thus, the outcome variable was imbalanced toward employees who remained in the organization.

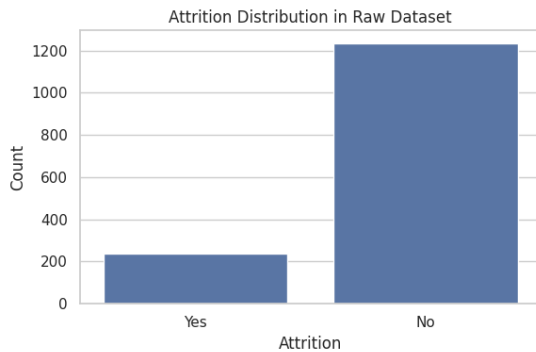


Fig. 1. Distribution of employee attrition in the complete dataset.

B. Hypothesis 1: There are significant differences in quantitative and ordinal employee metrics (e.g., age, monthly income, tenure, and distance from home) between employees who voluntarily resign and those who remain in the organization.

The first hypothesis examined whether selected numerical and ordinal employee characteristics differed between employees who left and those who remained. Mann–Whitney U tests were used because the analysis did not rely on a normality assumption for the compared variables. The resulting p -values were adjusted using the Benjamini–Hochberg FDR procedure.

After FDR correction, four characteristics showed statistically significant differences between the attrition groups: `MonthlyIncome`, `YearsInCurrentRole`, `StockOptionLevel`, and `DistanceFromHome`. `MonthlyIncome` showed the largest effect among the examined variables (rank-biserial $r = -0.322$), followed by `YearsInCurrentRole` ($r = -0.304$) and `StockOptionLevel` ($r = -0.264$). `DistanceFromHome` showed a smaller association in the opposite direction ($r = 0.126$).

Descriptive statistics from the complete dataset illustrate these differences. Employees who left had a median `MonthlyIncome` of 3,202, compared with 5,204 among employees who remained. Median `YearsInCurrentRole` was 2 years among employees who left and 3 years among employees who remained. In contrast, median `DistanceFromHome` was 9 among employees who left and 7 among those who remained. `NumCompaniesWorked`, `RelationshipSatisfaction`, `TrainingTimesLastYear`, and `PercentSalaryHike` did not show statistically significant differences after FDR correction. Therefore, the first hypothesis was partially supported: several, but not all, of the examined numerical and ordinal characteristics differed according to attrition status.

TABLE I
Mann–Whitney Tests for Numerical and Ordinal Characteristics

Variable	U	Raw p	FDR p	Rank-biserial r
<code>MonthlyIncome</code>	48555.5	< 0.001	< 0.001	-0.322
<code>YearsInCurrentRole</code>	49856.5	< 0.001	< 0.001	-0.304
<code>StockOptionLevel</code>	52757.5	< 0.001	< 0.001	-0.264
<code>DistanceFromHome</code>	80646.0	0.0099	0.0197	0.126
<code>NumCompaniesWorked</code>	77822.5	0.0702	0.0936	0.087
<code>RelationshipSatisfaction</code>	65250.5	0.0589	0.0936	-0.089
<code>TrainingTimesLastYear</code>	66974.0	0.1619	0.1850	-0.065
<code>PercentSalaryHike</code>	68518.5	0.3723	0.3723	-0.043

Figures 2 and 3 illustrate two of the strongest numerical differences observed in the univariate analysis.

C. Hypothesis 2: Categorical employee and workplace attributes (e.g., job role, overtime status, and stock option level) are significantly associated with employee attrition.

The second hypothesis examined whether categorical employee and workplace characteristics were associated with attrition. Pearson chi-square tests were applied to

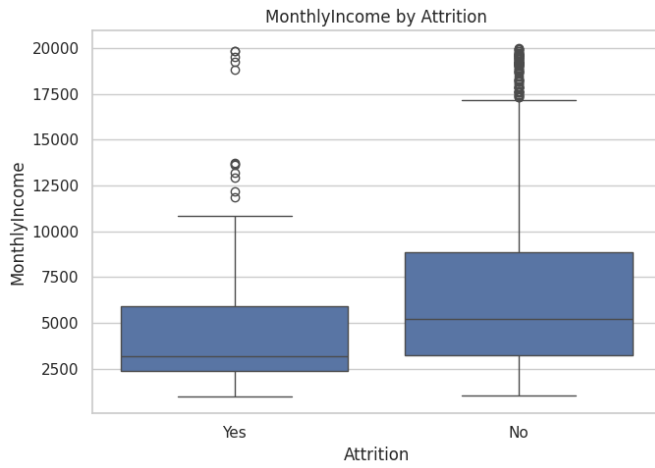


Fig. 2. Monthly income according to employee attrition status.

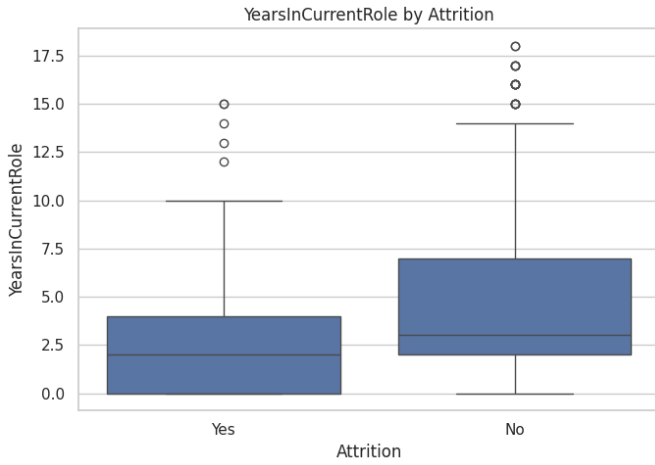


Fig. 3. Years in the current role according to employee attrition status.

the categorical variables, followed by Benjamini–Hochberg FDR correction.

The expected-frequency assumptions were generally satisfied. EducationField contained one expected cell below 5, corresponding to 8.33% of its contingency-table cells, with a minimum expected frequency of 2.90. All other examined categorical variables had no expected cells below 5.

After FDR correction, statistically significant associations with attrition were observed for JobRole, OverTime, MaritalStatus, BusinessTravel, Department, and EducationField. JobRole showed the largest association among the categorical characteristics (Cramer’s $V = 0.256$), followed by OverTime ($V = 0.211$) and MaritalStatus ($V = 0.175$). Gender was not significantly associated with attrition ($\chi^2(1) = 1.105$, FDR-adjusted $p = 0.293$).

The category-specific attrition rates further illustrate these associations. Employees who worked overtime had an attrition rate of 30.53%, compared with 10.44% among employees who did not work overtime. Employees who

traveled frequently had an attrition rate of 24.91%, whereas employees who did not travel had an attrition rate of 8.00%. Attrition also varied substantially across job roles and marital-status categories. These results support the second hypothesis by demonstrating associations between attrition and several categorical workplace and employee characteristics.

TABLE II
Chi-Square Tests for Categorical Characteristics

Variable	χ^2	df	FDR p	Cramer’s V
JobRole	75.173	8	< 0.001	0.256
OverTime	46.863	1	< 0.001	0.211
MaritalStatus	33.637	2	< 0.001	0.175
BusinessTravel	14.977	2	0.0010	0.112
Department	11.131	2	0.0054	0.094
EducationField	12.995	5	0.0273	0.088
Gender	1.105	1	0.2932	0.010

Figures 4 and 5 show the observed attrition patterns for two important workplace characteristics.

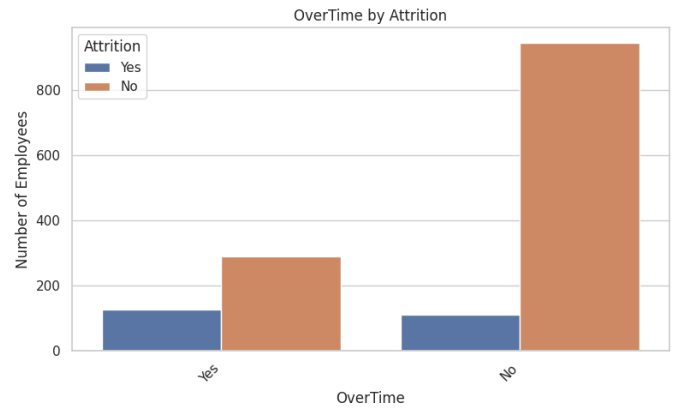


Fig. 4. Employee attrition according to overtime status.

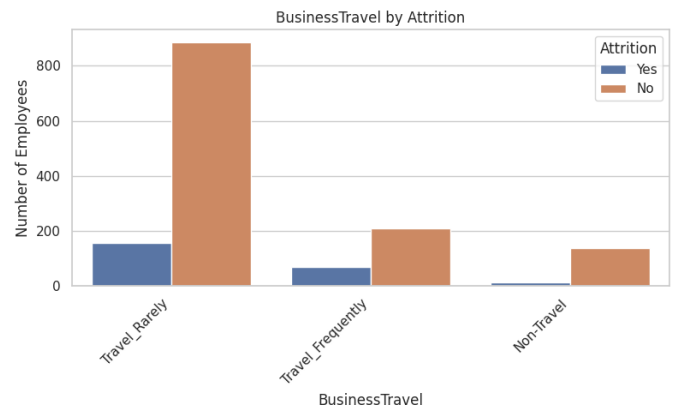


Fig. 5. Employee attrition according to business-travel frequency.

D. Hypothesis 3: Workplace and employee characteristics (specifically overtime work, job role, and stock option levels) remain independent predictors of employee attrition when controlling for confounding factors in a multivariable logistic regression model, with the effect of overtime being moderated by income level.

The third hypothesis examined which employee characteristics remained associated with attrition when considered simultaneously. A multivariable logistic regression model was fitted using the training data.

EducationField was evaluated as an additional group of terms using a generalized likelihood-ratio test (GLRT). Its inclusion did not significantly improve model fit ($LR = 6.535$, $df = 5$, $p = 0.258$), and EducationField was therefore not retained in the final model.

Two theoretically plausible interaction effects were subsequently examined. The interaction between OverTime and MonthlyIncome did not significantly improve model fit ($LR < 0.001$, $df = 1$, $p = 0.996$). Similarly, the interaction between DistanceFromHome and MaritalStatus was not significant ($LR = 0.684$, $df = 2$, $p = 0.710$). Neither interaction was therefore retained, resulting in a more parsimonious additive final model.

The final logistic regression model was statistically significant overall (likelihood-ratio $p < 0.001$) and had a McFadden pseudo- R^2 of 0.259. Several characteristics remained significantly associated with attrition after adjustment for the other variables in the model.

Employees who worked overtime had 4.54 times the odds of attrition compared with employees who did not work overtime (OR = 4.54, 95% CI [3.01, 6.83], $p < 0.001$). Frequent business travel was also strongly associated with attrition: employees who traveled frequently had 5.25 times the odds of attrition compared with employees who did not travel (OR = 5.25, 95% CI [2.18, 12.62], $p < 0.001$). Employees who traveled rarely also had higher odds than non-traveling employees (OR = 2.34, 95% CI [1.05, 5.22], $p = 0.038$).

Several job-role categories were also significantly associated with attrition relative to the model’s reference job role. Sales Representatives had substantially higher odds of attrition (OR = 14.63, 95% CI [4.08, 52.51], $p < 0.001$), as did Laboratory Technicians (OR = 7.44, 95% CI [2.40, 23.02], $p < 0.001$), Human Resources employees (OR = 5.09, 95% CI [1.34, 19.29], $p = 0.017$), and Sales Executives (OR = 5.04, 95% CI [1.78, 14.24], $p = 0.002$).

StockOptionLevel 1 and StockOptionLevel 2 were associated with lower odds of attrition relative to level 0 (OR = 0.204, 95% CI [0.105, 0.397], $p < 0.001$; and OR = 0.197, 95% CI [0.077, 0.504], $p = 0.001$, respectively). Each additional year in the employee’s current role was associated with lower odds of attrition (OR = 0.889, 95% CI [0.829, 0.953], $p < 0.001$). In contrast, each additional unit of DistanceFromHome (that is, every additional mile) was associated with higher odds of attrition (OR = 1.039, 95% CI [1.014, 1.063], $p = 0.002$), and each additional

company previously worked for was associated with higher odds (OR = 1.120, 95% CI [1.037, 1.209], $p = 0.004$).

An important difference emerged between the univariate and multivariable analyses. MonthlyIncome showed a statistically significant univariate association with attrition, but it was no longer significant after adjustment for the other characteristics in the logistic regression model (OR = 0.958 per 1,000 dollars, 95% CI [0.860, 1.068], $p = 0.442$). This result demonstrates that an association observed in isolation does not necessarily represent an independent association after other employee characteristics are taken into account.

TABLE III
Selected Significant Terms in the Final Logistic Regression Model

Predictor	OR	95% CI	p
OverTime: Yes	4.54	[3.01, 6.83]	< 0.001
Travel Frequently	5.25	[2.18, 12.62]	< 0.001
Travel Rarely	2.34	[1.05, 5.22]	0.038
Sales Representative	14.63	[4.08, 52.51]	< 0.001
Laboratory Technician	7.44	[2.40, 23.02]	< 0.001
Human Resources	5.09	[1.34, 19.29]	0.017
Sales Executive	5.04	[1.78, 14.24]	0.002
StockOptionLevel: 1	0.204	[0.105, 0.397]	< 0.001
StockOptionLevel: 2	0.197	[0.077, 0.504]	0.001
YearsInCurrentRole	0.889	[0.829, 0.953]	< 0.001
DistanceFromHome	1.039	[1.014, 1.063]	0.002
NumCompaniesWorked	1.120	[1.037, 1.209]	0.004

Overall, the third hypothesis was supported: several employee and workplace characteristics remained significantly associated with attrition after simultaneous adjustment, although some relationships observed in the univariate analysis, most notably MonthlyIncome, did not remain significant in the multivariable model.

E. Final Model Performance

The final model was evaluated on data not used for model fitting. On the validation set, the model achieved a ROC-AUC of 0.770 and a PR-AUC of 0.514. A classification threshold of 0.2044 was selected using the validation data and was subsequently applied unchanged to the independent test set.

On the test set, the final model achieved a ROC-AUC of 0.797 and a PR-AUC of 0.533. At the preselected threshold, test accuracy was 0.742, balanced accuracy was 0.723, recall for the attrition class was 0.694, precision was 0.352, and the F1 score was 0.467. Thus, the model identified approximately 69.4% of employees who left in the independent test sample, while the lower precision indicates that a substantial proportion of employees classified as likely to leave did not actually belong to the attrition class.

Figure 6 presents the ROC curve obtained from the final model predictions on the independent test set.

TABLE IV
Performance of the Final Logistic Regression Model

Metric	Validation	Test
ROC-AUC	0.770	0.797
PR-AUC	0.514	0.533
Threshold	0.204	0.204
Accuracy	0.800	0.742
Balanced Accuracy	0.757	0.723
Recall (Attrition)	0.686	0.694
F1 (Attrition)	0.527	0.467
Precision (Attrition)	0.429	0.352

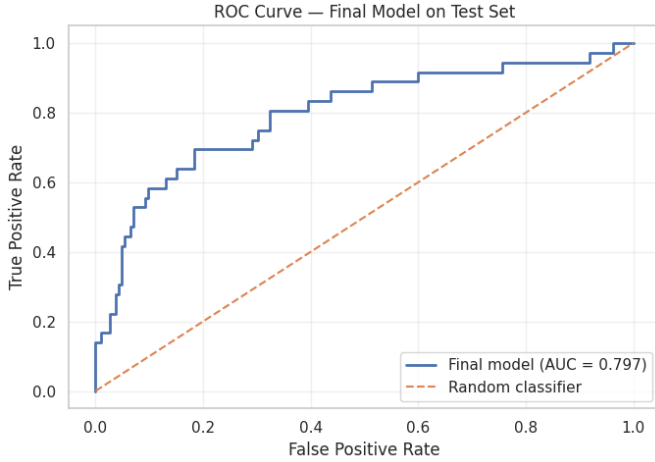


Fig. 6. Receiver operating characteristic curve of the final logistic regression model on the independent test set (ROC-AUC = 0.797).

III. Methods

A. Dataset and Preprocessing

The analysis was conducted using the IBM HR Analytics Employee Attrition dataset, which contains 1,470 employee observations and a range of demographic, compensation-related, career, and workplace characteristics. The outcome variable was Attrition, indicating whether an employee left the organization (Yes) or remained (No). For statistical modeling, this variable was encoded as a binary response, with attrition coded as 1 and non-attrition as 0.

Before statistical analysis, the dataset was inspected for data quality and variables with no analytical contribution (such as the Over18 variable which was uniformly 'Yes' for all employees) were removed where appropriate. Numerical and categorical variables were treated according to their measurement type. Monthly income was additionally expressed in thousands of dollars for the logistic regression model to improve coefficient interpretability without changing the underlying information.

To separate model development from final evaluation, the data were divided into training, validation, and test subsets using stratified sampling with respect to attrition status. Specifically, the dataset was partitioned into 70% training, 15% validation, and 15% test sets. Stratification

preserved approximately the same attrition proportion across the three subsets. The training set was used for statistical testing and model estimation, the validation set was used for classification-threshold selection and intermediate performance assessment, and the test set was reserved for a single final evaluation of the selected model.

B. Exploratory Analysis

Exploratory analysis was performed to characterize the dataset and examine the distributions of the variables before formal hypothesis testing. Summary statistics were calculated separately for employees who left and employees who remained. Numerical variables were examined using descriptive statistics and boxplots, while categorical variables were examined using frequency-based visualizations.

Potential redundancy among candidate numerical predictors was also examined before multivariable modeling. Correlations among the selected variables were inspected, and variance inflation factors (VIFs) were calculated using the training data to assess multicollinearity. These diagnostics were used to identify whether excessive linear dependence among predictors could compromise the stability or interpretation of the logistic regression coefficients. Predictors were selected based on theoretical relevance rather than automated Variance Inflation Factor (VIF) exclusion. This purposeful feature selection successfully mitigated initial multicollinearity, ensuring all variables in the final model achieved VIF scores well below the threshold of 5.

Figure 7 presents the correlation structure among the selected numerical variables considered prior to multivariable modeling. The correlation matrix was used as an exploratory diagnostic alongside the VIF analysis rather than as a formal hypothesis test.

C. Univariate Analysis of Numerical and Ordinal Variables

To evaluate whether continuous predictors followed a Gaussian distribution, visual inspections via Q-Q plots were conducted alongside formal Shapiro-Wilk test assessments. The Q-Q plots revealed noticeable tail deviations across most variables, and the Shapiro-Wilk tests yielded statistically significant results ($p < 0.01$) for all evaluated numerical features (see Table V). These findings confirm widespread departures from normality.

Because the normality assumption was violated across the continuous predictors, to evaluate whether selected numerical and ordinal employee characteristics differed between employees who left and those who remained, two-sided Mann-Whitney U tests were performed on the training set. This nonparametric procedure compares the distributions of two independent groups without requiring a normality assumption.

The null hypothesis for each tested characteristic was that its distribution did not differ between the attrition

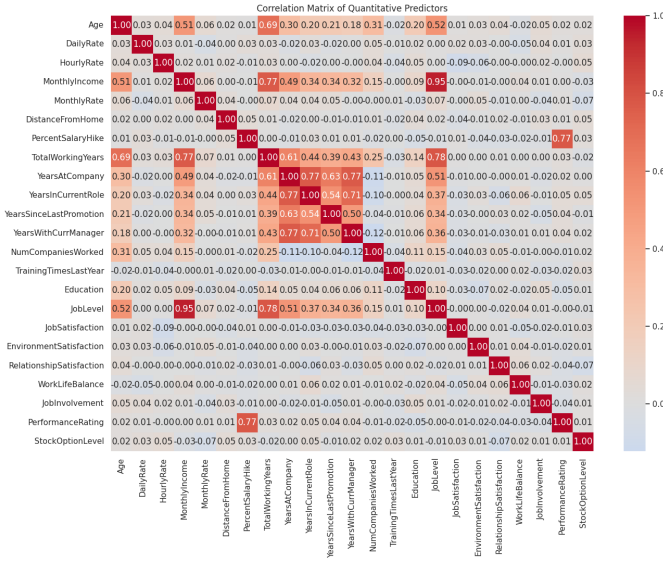


Fig. 7. Correlation matrix of the quantitative and ordinal predictors examined during exploratory analysis prior to multivariable modeling.

TABLE V
Shapiro-Wilk Test Results for Continuous Variables

Variable	Statistic (W)	p -value	Is Normal
Age	0.9786	0.0019	False
DailyRate	0.9441	0.0000	False
HourlyRate	0.9487	0.0000	False
MonthlyIncome	0.8129	0.0000	False
MonthlyRate	0.9584	0.0000	False
DistanceFromHome	0.8547	0.0000	False
PercentSalaryHike	0.8976	0.0000	False
TotalWorkingYears	0.9032	0.0000	False
YearsAtCompany	0.8212	0.0000	False
YearsInCurrentRole	0.8873	0.0000	False
YearsSinceLastPromotion	0.7219	0.0000	False
YearsWithCurrManager	0.8861	0.0000	False
NumCompaniesWorked	0.8775	0.0000	False
TrainingTimesLastYear	0.9054	0.0000	False

and non-attrition groups. The alternative hypothesis was that the distributions differed.

Because multiple numerical and ordinal characteristics were tested simultaneously, the resulting p -values were adjusted using the Benjamini–Hochberg false discovery rate (FDR) procedure. Statistical significance was evaluated at an FDR-adjusted significance level of $\alpha = 0.05$.

In addition to statistical significance, effect size was quantified using the rank-biserial correlation. For group 1 defined as employees who left and group 0 as employees who remained, it was calculated as

$$r_{rb} = \frac{2U}{n_1 n_0} - 1, \quad (1)$$

where U is the Mann–Whitney statistic and n_1 and n_0 are the sample sizes of the attrition and non-attrition groups, respectively. The sign indicates the direction of the group difference, while the magnitude reflects its strength.

D. Univariate Analysis of Categorical Variables

Associations between categorical employee characteristics and attrition were evaluated using Pearson chi-square tests of independence on the training set. For each characteristic, a contingency table was constructed between the variable categories and binary attrition status.

The Pearson chi-square statistic is given by

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad (2)$$

where O_{ij} and E_{ij} denote the observed and expected frequencies in cell (i, j) , respectively.

Expected cell frequencies were examined for every contingency table as a diagnostic of the chi-square approximation. Both the minimum expected frequency and the proportion of cells with expected frequencies below 5 were recorded. This allowed potential violations of the expected-frequency assumption to be identified rather than applying the test without diagnostic assessment.

As in the numerical analysis, multiple testing was addressed using the Benjamini–Hochberg FDR procedure. Correction was applied within the categorical family of tests, and an FDR-adjusted p -value below 0.05 was considered statistically significant.

The strength of each categorical association was additionally summarized using Cramer’s V :

$$V = \sqrt{\frac{\chi^2}{n \min(r-1, c-1)}}, \quad (3)$$

where n is the sample size and r and c are the numbers of rows and columns in the contingency table.

E. Multivariable Logistic Regression

A multivariable logistic regression model was used to examine the simultaneous association between selected employee characteristics and the probability of attrition. Unlike the univariate analyses, this model estimates the association of each predictor while adjusting for the other variables included in the model.

For employee i , the model can be expressed as

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \sum_{j=1}^k \beta_j X_{ij}, \quad (4)$$

where p_i is the probability of attrition, β_0 is the intercept, and β_j represents the coefficient associated with predictor X_j . Categorical predictors were represented using indicator variables with one category serving as the reference level.

Predictors for the multivariable analysis were specified as part of the research design rather than being selected automatically only because of their univariate p -values. This distinction allowed the univariate analyses to address the first two hypotheses while the logistic regression

addressed whether selected characteristics remained associated with attrition after simultaneous adjustment.

Regression coefficients were exponentiated to obtain odds ratios (ORs):

$$OR_j = e^{\beta_j}. \quad (5)$$

An odds ratio above 1 indicates higher estimated odds of attrition relative to the corresponding reference or unit increase, whereas an odds ratio below 1 indicates lower estimated odds. Confidence intervals for the odds ratios were obtained by exponentiating the corresponding coefficient confidence limits.

F. Likelihood-Ratio Tests and Interaction Assessment

Nested logistic regression models were compared using generalized likelihood-ratio tests (GLRTs). For a reduced model and a more complex full model, the likelihood-ratio statistic was calculated as

$$LR = 2(\ell_{\text{full}} - \ell_{\text{reduced}}), \quad (6)$$

where ℓ_{full} and ℓ_{reduced} are the maximized log-likelihoods of the two models. Under the null hypothesis, the statistic was compared with a chi-square distribution with degrees of freedom equal to the difference in the number of estimated model parameters.

This procedure was first used to evaluate whether the group of terms representing EducationField provided a significant improvement in model fit. The resulting reduced or full specification then served as the base model for subsequent interaction assessment, ensuring that the interaction models were constructed from the model selected in the preceding step.

Two prespecified interactions were investigated: OverTime $\times$ MonthlyIncome and DistanceFromHome $\times$ MaritalStatus. Each interaction model was compared with the corresponding additive base model using a GLRT. An interaction was retained only if its inclusion significantly improved model fit at $\alpha = 0.05$. This approach avoided increasing model complexity when the additional interaction terms were not supported by the data.

G. Model Validation and Classification Threshold

After the final logistic regression specification was determined using the training data, predicted attrition probabilities were calculated for the validation set. Model discrimination was evaluated using the area under the receiver operating characteristic curve (ROC-AUC) and the area under the precision-recall curve (PR-AUC).

Because the outcome was imbalanced and the default probability threshold of 0.5 was not assumed to be optimal for identifying attrition, the classification threshold was selected using the validation data rather than the test data. Candidate thresholds were evaluated on the validation set, and the selected threshold was subsequently fixed before final test-set evaluation.

Classification performance was summarized using accuracy, balanced accuracy, precision, recall, and the F1 score. In particular, balanced accuracy was included because ordinary accuracy can be misleading when one outcome class is substantially more frequent than the other.

H. Final Test Evaluation

The independent test set was not used for fitting the regression model, choosing its specification, or selecting the classification threshold. After all modeling decisions had been completed, the final model was applied once to the test data. Predicted probabilities were used to calculate ROC-AUC and PR-AUC, while binary predictions were obtained using the threshold previously selected from the validation set.

This separation between training, validation, and test data was used to reduce optimistic performance estimation and to provide an assessment of how well the final model generalized to observations not used during model development.

IV. Discussion

The present study examined employee attrition using complementary univariate and multivariable statistical approaches. Overall, the findings indicate that attrition is associated with a combination of workplace, career, and employee characteristics rather than with a single factor. The univariate analyses identified several differences between employees who left and those who remained, while the multivariable analysis demonstrated that some of these relationships persisted after simultaneous adjustment for other characteristics. Together, these findings provide support for the main research hypothesis that measurable employee and workplace characteristics are associated with attrition.

The first hypothesis was partially supported. MonthlyIncome, YearsInCurrentRole, StockOptionLevel, and DistanceFromHome differed significantly between the attrition groups after controlling the false discovery rate. The largest univariate effects were observed for MonthlyIncome and YearsInCurrentRole, with employees who left tending to have lower values for both. However, in the multivariable analysis, MonthlyIncome was no longer statistically significant after adjustment for the other predictors, whereas YearsInCurrentRole remained significant. This illustrates why univariate associations should not automatically be interpreted as independent effects: an observed relationship may partly reflect associations with other employee or workplace characteristics.

The second hypothesis was also supported for most categorical characteristics. JobRole, OverTime, MaritalStatus, BusinessTravel, Department, and EducationField were associated with attrition in the univariate analysis, whereas Gender was not. JobRole and OverTime showed the strongest associations. The substantially higher attrition

rate among employees working overtime was also reflected in the multivariable model, where overtime remained strongly associated with increased odds of attrition. BusinessTravel similarly remained important, particularly for employees who traveled frequently. These findings suggest that characteristics related to working conditions provide important information when examining attrition.

The third hypothesis was supported, with OverTime, BusinessTravel, selected JobRole categories, StockOptionLevel, YearsInCurrentRole, DistanceFromHome, and NumCompaniesWorked remaining significantly associated with attrition in the final logistic regression model. These findings are partly consistent with Yang and Islam [3] and suggest that attrition reflects multiple aspects of the employment experience. EducationField, despite being significant in the univariate analysis, did not improve the multivariable model. Likewise, neither the OverTime $\times$ MonthlyIncome nor DistanceFromHome $\times$ MaritalStatus interaction improved model fit, so the final model remained additive.

The final model achieved a test ROC-AUC of 0.797. At the validation-selected threshold, recall was 69.4% and precision was 35.2%, indicating that the model detected a substantial proportion of attrition cases but also produced a considerable number of false positives. These metrics should therefore be considered when applying the model to retention decisions.

Several limitations should be considered. First, the analysis is observational, so the estimated relationships represent statistical associations rather than causal effects. For example, the association between overtime and attrition does not establish that assigning overtime causes an employee to leave. Unmeasured characteristics may influence both predictors and the attrition outcome.

Second, the analysis uses a single IBM HR Analytics dataset containing 1,470 observations. Although separate training, validation, and test subsets protect against evaluating the model only on estimation data, an internal test set does not establish generalization to other organizations, industries, countries, or time periods. External validation on independent organizational datasets is therefore needed before drawing broader conclusions about predictive performance.

Third, the outcome was imbalanced, with substantially fewer employees leaving than remaining. This makes overall accuracy insufficient as a standalone performance measure and motivates evaluation of recall, precision, PR-AUC, and balanced accuracy. The relatively modest precision for the attrition class indicates that the model should not be interpreted as an error-free employee-level prediction system.

Fourth, some categorical regression estimates had relatively wide confidence intervals. For example, the estimated odds ratio for Sales Representatives was large but had a wide confidence interval, indicating considerable uncertainty about its exact magnitude. Such estimates

should therefore be interpreted mainly in terms of direction and uncertainty rather than the point estimate alone.

Finally, the analysis considered a predefined set of employee characteristics and two selected interaction structures. Other relevant variables or nonlinear relationships may exist but were not represented in the dataset or examined in the model. Future work could evaluate the findings using external datasets, investigate additional theoretically motivated interactions or nonlinear effects, and compare the interpretable logistic regression framework with alternative predictive approaches while maintaining strict separation between model development and final evaluation.

In conclusion, the analysis provides evidence that employee attrition is associated with multiple workplace and career-related characteristics. OverTime, BusinessTravel, JobRole, StockOptionLevel, YearsInCurrentRole, DistanceFromHome, and NumCompaniesWorked provided important information in the multivariable analysis, whereas some associations observed in isolation, such as MonthlyIncome, did not remain significant after adjustment. The results emphasize the value of examining attrition through both univariate and multivariable perspectives. At the same time, the observational nature of the data, uncertainty in some estimates, class imbalance, and lack of external validation require caution when translating these relationships into organizational decisions.

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